THE CEREBRAL CORTEX

REQUIRED READINGS: Power Point Presentation, these notes and pages of Blumenfeld Chapter 19 that are specified below

LEARNING OBJECTIVES: after studying this chapter, students should be able to:

1. Explain the anatomical organization of the cerebral cortex into neural layers and cortical columns
2. Describe the concept of Brodmann's areas
3. Discuss the concept of primary, secondary and association cortical areas
4. Explain the concept of lateralization and hemispheric dominance
5. Describe the location of the language centers and identify typical language deficits
6. Explain hemineglect syndrome and identify the cortical area involved in attention
7. Tell the location and function of the main areas for visual perception and the deficits produced by damage of those areas
8. Explain the function of the prefrontal association areas and the main deficits seen with damage of those areas
9. Define the different types of memory and name the 2 main features of the amnesic syndrome

Introduction

This chapter is dedicated to the study of the different regions and functions of the cerebral cortex. We will cover very briefly the layered and columnar organization and the function of the different layers. We will study the functional organization of the cerebral cortex in primary, secondary and association areas, and their unimodal and heteromodal functions as well as the concept of hemispheric dominance.

Required reading: Blumenfeld, Chapter 19.

Please, use these notes and the PP as a guide for your study.

OBJECTIVE 1: Anatomical organization of the cerebral cortex

Slide 3 & 4 - The cerebral cortex is about 2 to 5 mm thick with an estimated 100 billion neurons which are organized in 6 layers. The neocortex comprises most of the tissue commonly referred to as "the cortex". Practically all the areas seen from the outside are considered neocortex, evolutionary of a more recent acquisition. The 3 layered cortices in regions such as the hippocampus and olfactory cortices are evolutionary older areas, termed archicortex and paleocortex respectively.

The 6 layers of the neocortex are not similar everywhere. Neurons in the cortex are of different types and generally can be classified as pyramidal, fusiform and granular also called stellate. Pyramidal cells are abundant in areas of the cortex that originate long axons such as the motor cortex. These areas are named agranular cortices because small, granular neurons are not abundant. Other areas have only a few pyramidal cells and are dominated by small granular cells. These are called granular cortices. The pyramidal and fusiform cells give rise to most of the output axons from the cortex.

Slide 5 & 6 - Functionally the cerebral cortex is organized in vertical columns heavily interconnected up and down between cortical layers. These columns also connected horizontally with one another. Neurons in each column process the same information from layer 1 to the deepest cortical layer. The American Neuroscientist Vermont Mountcastle, studying the somatosensory cortex, was the first to show this columnar organization

Slide 7 - This slide reminds you about the columnar organization of the visual cortex. One area of the visual cortex which is named an ocular dominance column receives and analyzes information coming from one eye while the column next to it will do the same for the contralateral eye. Both columns processing information from the same point in space. This segregation of information is typical of all cortical areas.

OBJECTIVE 2 and 3: Neocortical areas can be divided according to their function

Slide 8 - Division into Brodmann's areas. (See Blumenfeld chapter 19, page 181 -183) A Brodmann area is a region of the cortex with a characteristic cytoarchitecture or cell organization. The German Neurologist and Neuroanatomist Korbinian Brodmann first defined them in 1909 studying brains from humans and other animals. Even if the concept of Brodmann areas has been debated extensively, they still are the most frequent way of naming the different cortical areas, mostly because they are not only cytoarchitecturally different but also correspond to different cortical functions.

Areas of the cortex are interconnected by fiber tracts:

- Association fibers – this name is given to the fiber tracts that connect different cortical areas within the same hemisphere.
- Commissural fibers – this name is given to the fiber tracts that connect homologous cortical areas in different hemispheres, such as the corpus callosum.
- Projection fibers – is the name given to the fiber tracts that connect cortical areas with other brain area such as brainstem, cerebellum etc.

Slides 9, 10 & 11 -

Each cerebral hemisphere is divided into primary and secondary sensory and motor areas:

- somatosensory cortex
- visual cortex

- auditory cortex
- vestibular cortex
- gustatory and olfactory cortex
- motor cortex

We call primary sensory areas the areas where sensory inputs coming from the periphery are first recognized and analyzed.

The primary motor areas contain the neurons (UMN) whose axons ultimately provide the descending pathways to influence the activity of LMN and movement. We talk about primary motor area or Brodmann's area 4; primary somatosensory areas or Brodmann's areas 1, 2 and 3; primary visual areas or Brodmann's area 17; primary auditory area or Brodmann's area 41 etc.

We refer to secondary, somatosensory, visual, auditory etc areas, as the cortical places receiving an input from the primary areas or the thalamus. These are called unimodal association areas. They are responsible for extracting more information out of the sensory input they receive. For example, they help with the interpretation of shape and texture of an object, color, light and direction of movement of an object, etc. Each new processing step will abstract new and more complex information.

The secondary motor areas (supplementary and premotor) function together with the primary motor cortex and the basal ganglia to generate patterns of motor movements.

There are also heteromodal association areas. Association areas are the areas that receive and analyze simultaneously input coming from motor and sensory cortices. They have bidirectional communications also with the thalamus, cerebellum, basal ganglia, limbic system, and some brainstem areas. This arrangement produces the highest and more complex mental functions that we know as "cognition". These cortices are prefrontal association area, the parieto-occipitotemporal junction or association area and the limbic association area.

Slides 12 – Synesthesia as an example of multisystem integration and processing

OBJECTIVE 4: The concept of hemispheric dominance or lateralization

Slide 13, 14 & 15

Even if anatomically the 2 cerebral hemispheres are almost identical, it has been known for a long time that each side is specialized in performing certain functions. Both hemispheres collaborate with each other through their corpus callosum connections. The reason for this different distribution of tasks is starting to be investigated. It has been known for a long time now that in infants the superior temporal area and the angular gyrus are larger in the left than in the right hemisphere and this could provide an explanation of the difference in task assignment. It is important to emphasize however that in most of the specialized functions both hemispheres work together and complement each other. We should think of lateralization and other complex cognitive functions more as an efficient subdivision of labor rather than a matter of superiority of one hemisphere over the other.

The best examples of lateralization of cortical function are:

- Handedness – See Blumenfeld Chapter 19, pages 883 – 885
- Language
- Attention

OBJECTIVE # 5:

- Language and language deficits - See Blumenfeld Chapter 19, pages 885 – 891

OBJECTIVE # 6

- Attention and spatial processing
- Attention deficits – See Blumenfeld Chapter 19, pages 898 – 901

OBJECTIVE # 7

- Areas for visual perception:
- Visual association cortex: See Blumenfeld Chapter 19, pages 913 – 917

OBJECTIVE # 8

- Functions of the prefrontal cortex – See Blumenfeld Chapter 19, page 906

OBJECTIVE # 9

- Memory disorders - Blumenfeld Chapter 19, page 838 - 842